

Scenario S3-O3-I1 — Our Level 3 (Fully Funded) / Investor Level 1 (Bootstrapped)

Scenario Summary

Structurally contradictory path. The founders over-invest at corporate intensity (~150-200% of ideal spend) through Pre-Pilot and Pilot 1, building a polished product and operational foundation — then a bootstrapped-level investor arrives right after Pilot 1 and contributes bare-minimum capital for scaling. The mismatch between the high-quality foundation and the under-funded scaling phase creates an awkward capital structure: excellent early execution followed by resource-starved growth.

- **Economic cost to Production:** \$150,000-190,000 / INR 12,450,000-15,770,000
- **Target timeline to Production:** 22-26 months
- **Founder/company capital before investor:** \$89,000-120,000 / INR 7,387,000-9,960,000
- **Investor capital modeled:** \$61,000-70,000 / INR 5,063,000-5,810,000
- **Equity structure:** 60 / 20 / 20 (founder / founding team / investor pool)
- **Main risk to watch:** high founder capital exposure followed by insufficient investor capital to sustain the scaling phases; the early over-spend creates expectations the bootstrapped investor cannot fund

Timeline

Phase	Timing	Duration	Minor Checkpoints	Gate to Advance
Pre-Pilot	M0-M3	3 months	Architecture freeze (M1), APK backlog lock and web hardening sprint (M1-M2), doctor training and admin SOPs complete (M2-M3), internal rehearsals at full intensity (M3).	Core web flows stable at high quality, internal rehearsals pass, doctor and admin runbooks versioned, APK build well advanced.
Pilot 1	M3-M5	2 months	Warm-network onboarding and first paid consults (M3-M4), callback calibration and doctor playbook iteration (M4), investor pitch deck finalized and outreach begins (M4-M5).	Ethiopia pilot runs live for 2 months with measured throughput; investor conversations active.
Pilot 2	M5-M12	7 months	Investor close (M5-M6), Kenya prep (M8), partner reporting (M9), roster planning and partial automation (M10), staffing calibration (M11-M12).	5k-10k user-capable operations with stronger APK, cleaner reporting, staffed callback coverage.
Production Readiness	M12-M22	10 months	Compliance closeout (M14), release freeze (M16), launch playbooks (M18), certification refreshers (M19), production support schedule (M20-M22).	Current prototype scope is production-safe across web and patient mobile with documented fallback operations.

Feature and Output Checklist

Phase	Output Checklist
Pre-Pilot	Web core hardened at full intensity; APK build advanced (parallel with web work); doctor/admin training materials complete; support desk scripts and callback trees defined and rehearsed. Fast iteration due to high utilization.
Pilot 1	Patient app covers broad journeys (registration, dashboard, consult, prescriptions, labs); provider web stable for GP and specialist flows; doctor roster, admin callbacks, and manual fulfilment visible in-system. High-quality output from fully funded Pre-Pilot.
Pilot 2	Core patient features complete but feature expansion slows due to bootstrapped investor capital; partner dashboards functional; staffing ladder active at minimum levels; reporting and payment loops operational but constrained. Kenya prep begins late in phase.
Production Readiness	Current prototype scope is production-grade across web and patient mobile. Trained ops staff with manual fallback available. Compliance documentation complete. Extended Production Readiness phase compensates for under-funded Pilot 2.

Services and SLAs

Phase	Uptime Target	Response Commitment	Operating Model
Pre-Pilot	95.5%	Same business day	3 core team at full utilization + 7 part-time developers at 150% capacity (640 hrs/mo) + 2 advisors; no employed clinical coverage, only training and rehearsals.
Pilot 1	97.0%	Same-day for critical issues	Minimum employed coverage: 3 doctors, 3 admin ops, 2 technical support. Partner clinics supplement. Higher quality from fully funded build.
Pilot 2	97.5%	4-hour P1 response	Minimum employed coverage drops to constrained levels: 3 doctors, 3 admin ops, 2 technical support. Bootstrapped investor capital limits staff expansion.
Production	98.5%	2-hour P1 response	Minimum employed coverage: 5 doctors, 5 admin ops, 3 technical support. Slightly below ideal due to capital constraints.

Cost Breakdown

Summary

Category	Amount
Capital expenditure through Production	\$82,000 / INR 6,806,000
Operating expenditure before Production launch	\$88,000 / INR 7,304,000
Total economic spend to Production	\$170,000 / INR 14,110,000

Workstream Detail

Workstream	Economic Cost (USD)	Economic Cost (INR)	Notes
Product and founder office	\$39,600	INR 3,286,800	3 core team at \$20/hr x 88 hrs/mo x 150% utilization = ~\$7,920/mo for Pre-Pilot+Pilot 1 (5 months), then drops to ~\$5,280/mo post-investor (investor level constrains pace).
Web platform hardening	\$14,900	INR 1,236,700	745 hours at \$20/hr senior rate. Compressed into Pre-Pilot and Pilot 1 at high intensity.
Patient APK build	\$7,200	INR 597,600	Fixed Android quote: 480 hours at \$15/hr. Parallelized with web work during fully funded Pre-Pilot.
QA, DevOps, security	\$10,400	INR 863,200	~520 hours at \$20/hr. Fully funded Pre-Pilot allows thorough hardening.
Admin and care operations	\$62,400	INR 5,179,200	Employed doctors, admin callback staff, and technical support across 19 months of live operations. Constrained by bootstrapped investor during Pilot 2 and Production Readiness.
Partner onboarding and provider success	\$9,500	INR 788,500	Doctor/admin training protocols, SOP writing, partner reporting. Well-executed during Pre-Pilot.
Legal, accounting, compliance	\$4,200	INR 348,600	Shorter timeline than S2-O1-I1 but still requires all compliance touch-points.
Infrastructure, phones, and tools	\$19,800	INR 1,643,400	Hosting, devices, telecom/call costs across 22 months. Higher during Pre-Pilot (fully funded), drops during investor-constrained phases.
Marketing, travel, rehearsals	\$6,000	INR 498,000	Moderate pre-investor, lean post-investor.
Contingency and buffer	\$26,000	INR 2,158,000	~15% buffer. Lower than S2-O1-I1 because the fully funded Pre-Pilot reduces early-phase risk.

Operational Staffing Cost by Phase

Phase	Duration	Monthly Ops Cost (USD)	Monthly Ops Cost (INR)	Phase Total (USD)
Pre-Pilot (M0-M3)	3 months	\$0	INR 0	\$0
Pilot 1 (M3-M5)	2 months	\$3,200	INR 265,600	\$6,400
Pilot 2 (M5-M12)	7 months	\$3,100	INR 257,300	\$21,700
Production Readiness (M12-M22)	10 months	\$5,200	INR 431,600	\$52,000
Total operational staffing				\$80,100

Infrastructure Cost by Phase

Phase	Duration	Monthly Infra (USD)	Monthly Infra (INR)	Phase Total (USD)
Pre-Pilot	3 months	\$130	INR 10,790	\$390
Pilot 1	2 months	\$450	INR 37,350	\$900
Pilot 2	7 months	\$500	INR 41,500	\$3,500
Production Readiness	10 months	\$800	INR 66,400	\$8,000
One-time equipment (phones, laptops)	—	—	—	\$1,400
Total infrastructure				\$14,190

Effort and Team

Function	Staffing Pattern	Notes
Core team	3 part-time members @ ~4 hrs/day each, ~150% effective utilization pre-investor	Single owner/founder + 2 founding members. High intensity during Pre-Pilot and Pilot 1.
Technical delivery (pre-investor)	7 part-time developers, modeled at ~640 productive hrs/mo	Full utilization. Maximum parallelism between Android and web.
Technical delivery (post-investor)	7 part-time developers, modeled at ~240 productive hrs/mo	Investor capital at bootstrapped level constrains team utilization to ~50% of max. Significant drop from pre-investor pace.
Advisory board	2 fractional advisors @ ~10-15 hrs/mo each	Finance, compliance, strategic review.
Employed care coverage	Pilot 1: 3 doctors / Pilot 2: 3 / Production: 5	Below ideal due to bootstrapped investor. Partner clinics critical.
Employed admin ops	Pilot 1: 3 admins / Pilot 2: 3 / Production: 5	Callbacks, exception handling, manual fulfilment. Constrained post-investor.
Technical support	Pilot 1: 2 / Pilot 2: 2 / Production: 3	L1/L2 issue triage and live-ops monitoring.

Monthly Team Economic Cost

Period	Dev Team (USD)	Dev Team (INR)	Notes
Pre-investor (M0-M5)	~\$22,245	~INR 1,846,335	~150% of full \$14,830/mo rate card. Full corporate intensity.

Post-investor (M6-M22)	~\$7,400	~INR 614,200	~50% of full rate card. Bootstrapped investor forces sharp reduction.
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Revenue and Pricing

Revenue Source	Pilot 1	Pilot 2	Production
GP consults	Subsidised, Ethiopia-first, ~\$2-3 per paid interaction	Full Ethiopia pricing + Kenya tests	Core revenue line
Specialist consults	Some activity (fully funded Pilot 1 enables specialist flows)	Grows slowly after referral flow stabilizes	Lower-volume margin contributor
Pharmacy commission	Small, admin-assisted	Repeatable, manual-heavy	Scales with partner conversion
Diagnostics commission	Small, admin-assisted	Improves with order flow consistency	Retention lever
Travel/care coordination	Rare, high-touch	Low-volume, higher-value	Strategic differentiator

Revenue Line	Ethiopia Benchmark	Kenya Benchmark	Notes
GP consult	\$3.5 / ETB 470	\$5 / KES 650	Starts subsidised in Pilot 1; reaches benchmark by late Production.
Specialist consult	\$9 / ETB 1,250	\$12 / KES 1,560	Pilot 2 onward.
Pharmacy take-rate	10% of order value	10% of order value	Pilot 1 onward with manual fulfilment.
Diagnostics take-rate	12% of order value	12% of order value	Pilot 1 onward with admin/lab coordination.
Travel/care coordination	\$10-15	\$10-15	Small volume, high-touch.
Subscription	Deferred	Deferred	Not modeled before meaningful scale.

CAC and Key Business Metrics

Metric	Pilot 1	Pilot 2	Production
CAC	\$8	\$14	\$20
Gross margin	40%	48%	55%
Monthly ARPU	\$2.0	\$3.0	\$4.5
Monthly churn	12%	10%	8%
LTV/CAC target	1.5x	2.2x	3.0x

Cash Flow and Runway

Phase Summary

Checkpoint	Ending Cash (USD)	Ending Cash (INR)	Approx. Runway Remaining	Trigger Threshold
Pre-Pilot (M3)	\$5,200	INR 431,600	0.5 months	Critical: very thin runway due to high Pre-Pilot spend. Investor must be identified.
Pilot 1 (M5)	\$3,800	INR 315,400	0.3 months	Extremely tight. Investor close must happen by M6.
Pilot 2 (M12)	\$28,900	INR 2,398,700	2.8 months	Trigger contingency if cash drops below 2 months of burn.
Production Readiness (M22)	\$16,500	INR 1,369,500	1.8 months	Trigger contingency if cash drops below 2 months of burn.

Monthly Cash Flow

Month	Phase	Founder Capital In	Investor In	Revenue In	Outflow	Net	Ending Cash
M1	Pre-Pilot	\$22,500	\$0	\$0	\$22,375	\$125	\$125
M2	Pre-Pilot	\$22,500	\$0	\$0	\$22,375	\$125	\$250
M3	Pre-Pilot	\$22,500	\$0	\$0	\$22,375	\$125	\$375
M4	Pilot 1	\$22,500	\$0	\$300	\$25,575	-\$2,775	-\$2,400
M5	Pilot 1	\$22,500	\$0	\$600	\$25,575	-\$2,475	-\$4,875
M6	Pilot 2	\$0	\$61,000	\$1,000	\$10,500	\$51,500	\$46,625
M7	Pilot 2	\$0	\$0	\$1,400	\$10,500	-\$9,100	\$37,525
M8	Pilot 2	\$0	\$0	\$1,800	\$10,500	-\$8,700	\$28,825
M9	Pilot 2	\$0	\$0	\$2,200	\$10,500	-\$8,300	\$20,525
M10	Pilot 2	\$0	\$0	\$2,600	\$10,500	-\$7,900	\$12,625
M11	Pilot 2	\$0	\$0	\$3,000	\$10,500	-\$7,500	\$5,125

M12	Pilot 2	\$0	\$0	\$3,400	\$10,500	-\$7,100	-\$1,975
M13	Prod. Readiness	\$0	\$0	\$3,800	\$12,600	-\$8,800	-\$10,775
M14	Prod. Readiness	\$0	\$0	\$4,400	\$12,600	-\$8,200	-\$18,975
M15	Prod. Readiness	\$0	\$0	\$5,000	\$12,600	-\$7,600	-\$26,575
M16	Prod. Readiness	\$0	\$0	\$5,800	\$12,600	-\$6,800	-\$33,375
M17	Prod. Readiness	\$0	\$0	\$6,500	\$12,600	-\$6,100	-\$39,475
M18	Prod. Readiness	\$0	\$0	\$7,500	\$12,600	-\$5,100	-\$44,575
M19	Prod. Readiness	\$0	\$0	\$8,500	\$12,600	-\$4,100	-\$48,675
M20	Prod. Readiness	\$0	\$0	\$10,000	\$12,600	-\$2,600	-\$51,275
M21	Prod. Readiness	\$0	\$0	\$11,500	\$12,600	-\$1,100	-\$52,375
M22	Prod. Readiness	\$0	\$0	\$13,000	\$12,600	\$400	-\$51,975

Totals:

- Founder capital in: \$112,500 / INR 9,337,500 (M1-M5 at ~\$22,500/mo x 5 months)
- Investor capital in: \$61,000 / INR 5,063,000
- Cumulative revenue: ~\$91,400
- Cumulative outflow: ~\$316,875

WARNING: Cash goes negative at M12 and continues deepening through M22. The bootstrapped investor provides insufficient capital to sustain Pilot 2 and Production Readiness at the pace established during the fully funded Pre-Pilot. Either additional capital injection is required during Production Readiness, or the ops team must be cut — undermining the fully funded foundation.

ROI and Break-Even

Metric	Value
Total capital invested (founder + investor)	\$173,500 / INR 14,400,500
Total economic cost	\$170,000 / INR 14,110,000
Sweat-equity gap (economic cost minus cash invested)	~\$0 (fully funded Pre-Pilot minimizes gap)
Cumulative revenue at M22	~\$91,400 / INR 7,586,200
Monthly break-even (revenue >= outflow)	~M22 (month 22)
Cumulative break-even (total revenue >= total capital)	~M30-M36 (8-14 months post-Production)
Time to ROI (revenue exceeds all economic cost)	~M34-M40 (12-18 months post-Production)

Fundraising, Valuation, and Dilution

Item	Value
Founder/self-funded capital before investor	\$112,500 / INR 9,337,500
Investor round size	\$61,000 / INR 5,063,000
Recommended structure	SAFE or milestone-based bridge note
Modeled pre-money valuation	\$305,000
Modeled post-money valuation	\$366,000
Modeled investor dilution from the 20% pool	16.7%

Ownership Pool	Before Round	After Modeled Round	Notes
Owner / founder pool	60%	56.0%	Retained majority control.
Founding + strategic pool	20%	18.7%	Shared pool for founding members and strategic partners.
Investor pool	20% reserved	16.7% issued / 3.3% remaining	Drawn from the dedicated investor pool. Bootstrapped investor takes smaller slice.

Key Risks

Risk	Likelihood	Impact	Mitigation / Trigger
Capital structure mismatch: fully funded early, under-funded late	Very High	Critical	The defining risk of this scenario. The high-quality Pre-Pilot and Pilot 1 set expectations that the bootstrapped investor cannot sustain. Either seek additional investor capital or accept quality regression during Pilot 2.
Cash exhaustion during Pilot 2 / Production Readiness	Very High	Critical	Cash goes negative by M12. Must either find supplementary capital, cut ops staff, or accept the path is not viable.

Founder over-exposure (\$112,500 in 5 months)	High	High	\$22,500/mo for 5 months requires substantial personal reserves. If Pilot 1 does not attract an investor, the founder has significant sunk cost.
Bootstrapped investor adds insufficient value	High	High	\$61,000 is barely enough for 7 months of Pilot 2. Production Readiness is entirely revenue-dependent.
Team morale whiplash	Medium	High	Moving from corporate-intensity Pre-Pilot to bootstrapped scaling creates jarring adjustment. Team may disengage.
Regulatory delay (Ethiopia / Kenya)	Medium	High	Continue partner-license model. Compliance work front-loaded in Pre-Pilot.
Manual ops overload during Production Readiness	High	Medium	Understaffed ops team cannot support growing user base. Add admin capacity before feature expansion.
Payment-rail mismatch	Medium	Medium	Use digital invoicing first; localize M-Pesa / Telebirr before scaling.

Recommendation

S3-O3-I1 is **structurally contradictory and should be avoided**. The founders over-invest at corporate intensity (\$89,000-120,000 over 3-5 months), building a high-quality foundation — then a bootstrapped investor contributes only \$61,000-70,000, which is insufficient to sustain the operating level established during the fully funded early phases.

This creates a predictable failure mode:

1. The Pre-Pilot and Pilot 1 are excellent — high utilization, thorough hardening, comprehensive training
2. The investor arrives with bare-minimum capital
3. The team must dramatically reduce intensity to match the new capital constraint
4. The quality regression damages team morale, partner confidence, and operational consistency
5. Cash goes negative during Pilot 2, requiring emergency capital or scope cuts

This path should only be considered if:

1. The founders view the fully funded Pre-Pilot as an investment in product quality that will attract a better investor later (treating I1 as a bridge to a future I2/I3 round)
2. The bootstrapped investor brings strategic value (network, partnerships, market access) beyond their capital contribution
3. Revenue ramps faster than modeled, compensating for the under-funded investor

In nearly all cases, S3-O2-I2 or S3-O3-I2 would be superior alternatives — the first reduces founder exposure while properly funding scaling, and the second maintains the fully funded foundation with adequate investor capital.

Assumption Notes

- Android is modeled from the external quote: 480 total hours at \$15/hr average across the mobile team.
- Remaining web work is normalized to 150%-160% of Android effort, with an expected case of 745 hours.
- Salary benchmarks for employed coverage: INR 60,000 for doctors (\$720/mo), INR 18,000 for admin ops (\$217/mo), INR 25,000 for technical support (\$301/mo).
- Our L3 (fully funded) spend is modeled at ~150% of ideal team utilization, yielding ~\$22,500/mo founder contribution during Pre-Pilot and Pilot 1.
- Investor L1 (bootstrapped) contribution sized at ~\$61,000-70,000, representing bare-minimum capital for post-Pilot 1 phases.
- S3 timing: investor arrives right after Pilot 1 completes (M5-M6).
- All INR conversions at 1 USD = 83 INR per params.md.
- FX buffers of 10-15% applied to ETB and KES revenue projections.

End of S3-O3-I1 scenario. All assumptions trace to params.md v2.